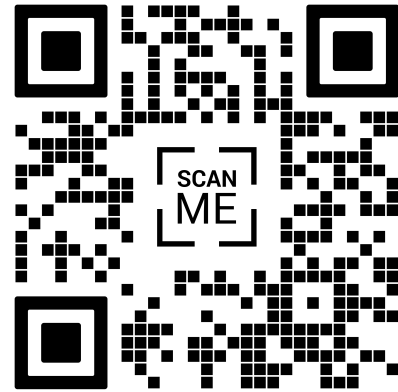




SINA TAVAKOLI

Data Scientist ~ Data Engineer

 [LinkedIn](#)
 [GitHub](#)
 [Kaggle](#)

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ABOUT ME

Data Scientist and Data engineer with over four years of experience specializing in data analytics, machine learning, and scalable Python solutions. Currently pursuing a Master's in ICT, with proven success in optimizing data processing workflows, achieving up to 30% improvement in model accuracy, and reducing processing times by 20%. Skilled in utilizing Python, SQL, and machine learning frameworks (e.g., PyTorch, TensorFlow) to drive data-driven decisions and support predictive analytics. Seeking to leverage expertise in ML model development and data pipelines to contribute to innovative projects in a dynamic environment. A proactive problem-solver with a strong focus on innovation, efficiency, and quality.

SKILLS

Programming Languages: Python, C++, MATLAB, R
Concepts: OOP, Software Development Lifecycle, IT Service Management, Data Pipelines, Big Data Processing, Data Engineering, Cloud Platforms (AWS, GCP, Azure)
Libraries/Tools: PyTorch, Scikit-Learn, Pandas, Numpy, OpenCV, TensorFlow, Git, MySQL, Hugging Face, dbt
Data Engineering Tools: Apache Airflow, Kestra, Terraform, Docker, ETL, Data Workflow Orchestration, Data Ingestion
Architectures: CNN, RNN, LSTM, Transformers (BERT, RoBERTa, ALBERT, DistilBERT)
Additional: API Integration, Data Visualization, Data Cleaning & Preprocessing, Web Scraping, Database Management, Version Control

PERSONAL SKILLS

- Communicates complex technical concepts effectively to diverse audiences, ensuring understanding across technical and non-technical stakeholders.
- Employs analytical thinking and structured approaches to identify innovative solutions for project challenges.
- Consistently adopts new methodologies and technologies to enhance project outcomes and support team goals.
- Collaborative and dependable team contributor, positively impacting project dynamics and driving successful results.

EDUCATION

2024 - Present	Master of Business Informatics, Mobility Program Participating in a structured mobility program as part of my Master's studies, focusing on business informatics and its applications in data-driven decision-making, enterprise systems, and digital transformation. This program enhances my expertise in integrating business and IT strategies, equipping me with analytical and technical skills essential for data science and engineering roles.	Mannheim University
2022 - 2024	Master of ICT (Information and Communication Technology) Master's Thesis: <i>Textual Embeddings and Neural Networks for Emotion and Slur Prediction: A Comparative Study of BERT Variants</i> Engineered a classification framework using BERT-based embeddings to detect emotions and slurs in textual data. The research employed multiple BERT variants and evaluated their effectiveness, contributing to sentiment analysis and content moderation.	Università degli Studi di Padova
2014 - 2019	Bachelor of Electrical Engineering Bachelor's Thesis: <i>Energy Consumption Optimization in Smart Grids Using Data Analytics</i> Designed a predictive model for optimizing energy consumption in smart grid systems by analyzing consumption patterns and forecasting demand, refining energy efficiency by minimizing wastage and balancing supply, aiding sustainable energy management.	Razi University

EXPERIENCE

09/2023–Current	Teaching Assistant	Università degli Studi di Padova
	• Led tutorials and lab sessions on network analysis using Python and R, designing hands-on activities that elevated student scores by 20%. • Designed data-driven assignments using Python's data manipulation libraries, delivering a 90% student satisfaction rate through interactive learning methods. • Provided mentorship in Python and R programming, helping students increase their modeling accuracy by an average of 15%.	
03/2021–07/2022	Data Scientist	Program and Budget Organization
	• Processed and managed large datasets (up to 50 GB), achieving a 25% reduction in processing time by optimizing data structures with Pandas and NumPy. • Engineered predictive models for budget forecasting using SciPy and StatsModels, enhancing forecast accuracy by 30%.	
06/2018–03/2021	Python Developer	Fara Sanat
	• Contributed to Python applications for customer satisfaction analysis, adding key functionalities to improve project outcomes. • Executed unit testing and code reviews using PyTest, decreasing bug occurrences by 20%. • Built scalable Python applications, increasing efficiency by 50% using advanced algorithms.	

PROJECTS

- **Abstract Base Sentiment Analysis** ([GitHub Link](#))
Developed and optimized a slur classification model using BERT-based transformers, achieving 90% accuracy. Improved data preparation techniques, reducing error rates by 15%.
- **Breast Cancer Image Classification** ([GitHub Link](#))
Created a classification model for breast cancer, accurately categorizing images as benign, malignant, or normal, with a 90% accuracy rate using advanced feature extraction.
- **Customer Segmentation Analysis** ([GitHub Link](#))
Applied K-means clustering for customer segmentation, generating insights that supported three product launches within six months.
- **Analyzing Ukraine War** ([GitHub Link](#))
Analyzed Ukraine war discussions on Reddit with Python, generating visualizations and insights on community engagement.
- **Twitter's Impact on the Olympics** ([GitHub Link](#))
Investigated Twitter data on the Olympics, achieving 85% accuracy in sentiment and topic analysis.
- **Corrupted Image and Patches** ([GitHub Link](#))
Built a C++ application to restore corrupted image regions, reducing manual restoration time significantly.

CERTIFICATES

- **Basic Python Programming Certificate** (57 hours)
- **Advanced Python Programming Certificate** (57 hours)
- **GitHub Certificate** (30 hours)
- **Introduction to Algorithms and Programming** (60 hours)
- **Data Science Certificate** (60 hours)

LANGUAGES

English - C1, Italian - A1, German - A2, Persian - Native